QR Decomposition:

$A = QR$ where $Q = [\vec{q_1} \cdots \vec{q_n}]$ when with $\vec{q_i} \cdot \vec{q_j} = 0$

$\& \ \vec{q_i} \cdot \vec{q_i} = 1$

$R \rightarrow$ upper triangular matrix.

we! Gram schmidt approach: given $\vec{a_1}, \cdots \vec{a_n}$ from A

gen $\vec{q_1}, \cdots \vec{q_n}$.

① $\vec{q_1} = \frac{\vec{a_1}}{\|\vec{a_1}\|} = \frac{\vec{a_1}}{r_{11}}$ where $r_{11} = \|\vec{a}\|$ $\Rightarrow \vec{a_1} = r_{11}\vec{q_1}$

② $\vec{b_2} = \vec{a_2} - \text{proj}_{\vec{q_1}} \vec{a_2} = \vec{a_2} - (\vec{a_2} \cdot \vec{q_1})\vec{q_1}$;

$\Rightarrow \vec{a_2} = r_{12}\vec{q_2} + (\vec{a_2} \cdot \vec{q_1})\vec{q_1}$; $\vec{q_2} = \frac{\vec{b_2}}{\|\vec{b_2}\|} = \frac{\vec{b_2}}{r_{22}}$.

③ $\vec{b_k} = \vec{a_k} - \sum_{i=1}^{k-1} \text{proj}_{\vec{q_i}} \vec{a_k} = \vec{a_k} - (\vec{a_k} \cdot \vec{q_i})\vec{q_i}$,

$\vec{a_k} = r_{kk}\vec{q_k} + \sum_{i=1}^{k-1} \langle \vec{a_k}, \vec{q_i} \rangle \vec{q_i}$ $\qquad \vec{q_k} = \frac{\vec{b_k}}{r_{kk}}$.

One approach: ~~we~~ use gram schmidt to get $\vec{q_1}, \vec{q_2}, \cdots \vec{q_n}$

from $\vec{a_1}, \cdots \vec{a_n}$ & write columns of A

as linear comb of $\vec{q_1}, \cdots, \vec{q_n}$.

$\vec{r_{ij}}$: $Q\vec{r_{ij}} = \vec{a_i}$

$Q\vec{x} = \vec{b}$ where $Q = [\vec{a_1}, \ldots \vec{a_n}]$ orthogonal & unitnorm

solving for $\vec{x}$ → computing coeff of LC of

$$\sum x_i \vec{Q_i} = \vec{b}$$

since $(\vec{Q_1}, \ldots, \vec{Q_n})$ are orthogonal & unitnorm

$$\Rightarrow x_i = \vec{b} \cdot \vec{Q_i}$$

$$\boxed{\vec{x} = Q^T \vec{b}}$$

$$\boxed{\vec{q_i} = Q^T \vec{a_i}}$$

$$\boxed{\vec{a_k} = \vec{Q_{1c}} q_{1k} + \sum_{i=1}^{k-1} (\vec{a_k} \cdot \vec{Q_i}) \vec{Q_i}}$$

$$= \sum_{i=1}^{k} x_i \vec{Q_i}$$

where $x_i = (\vec{a_k} \cdot Q_i)$ $\forall i=1$

$$\boxed{x_k = \|\vec{b_k}\|}$$

every column vector of A

= Coeff of LC of columns of Q

I

linear combination

k^{th} column vector of A = linear comb of first k columns of Q

$$[\vec{a_1}, \ldots, \vec{a_n}] = [\vec{q_1} \ldots \vec{q_n}] \begin{bmatrix} \vec{r_1}, & \cdots & \vec{r_n} \end{bmatrix}$$

$$= [\vec{q_1}, \ldots \vec{q_n}] \begin{bmatrix} r_{11} & r_{12}\vec{q_1} & a_3\vec{q_2} \\ 0 & r_{22} & \vec{a_3}\vec{q_2} \\ \vdots & 0 & r_{33} \\ 0 & 0 & \ddots \end{bmatrix}$$

upper triangular matrix.

$$= QR$$

where $\vec{b}$ keeps changing

$A\vec{x} = \vec{b}$

$(QR)\vec{x} = \vec{b}$

$Q^T QR\vec{x} = Q^T\vec{b}$

Since $Q^T = Q^T$ from before

$R\vec{x} = Q^T\vec{b}$

Existance of solution for over determined sys of linear eqns.

(if eqns map to subspaces)

Given $A\vec{x}=\vec{b}$ where $A \in \mathbb{R}^{m \times n}$ with $m > n$

$\vec{x} \in \mathbb{R}^n$ and $\vec{b} \in \mathbb{R}^m$

sol^n exists if $\vec{b} \in \text{col}(A) \subset \mathbb{R}^m$

↳ cannot span $\mathbb{R}^m$ since $\vec{x}$ vectors of length n.

$\Downarrow$

$\exists\, \vec{b} \notin \text{col}(A)$

↳ $\vec{b} \in \mathbb{R}^m$

if $\vec{b} \in \text{col}(A)$ → solve to determine $\vec{x}$: $A\vec{x}=\vec{b}$

if $\vec{b} \notin \text{col}(A)$ → find an Approximate sol: (lies in the col(A) $\ni \hat{b}$)

$\hat{x}$ $\quad A\hat{x}=\hat{b}$

$\|\hat{b}-\vec{b}\|$ is minimum.

Approximal sol: $\hat{x}$

① come up with $\hat{b} \in \text{col}(A)$ s.t. $\|\hat{b}-\vec{b}\|$ is minimum.

② Solve $A\hat{x}=\hat{b}$ ↑ approximate sol.

when $\vec{b} \in \text{col}(A)$, $\hat{x}$ is an exact sol.

finding $\hat{b}$

col

Spa

Recall:

$A^T(\vec{b}-\hat{b})$

↓

Zero

finding $\hat{b}$: proj$_{\text{col}(A)}$ $\vec{b}$

Column vectors of A i.e $\{\vec{a_1}, \ldots \vec{a_n}\}$ is a spanning set of col(A)

Recall: if $\hat{b}$ is a proj of $\vec{b}$ on to col(A), $\vec{b}-\hat{b} \perp$ every vector in col(A)

$\Rightarrow$ $\vec{a_1}^T(\vec{b}-\hat{b}) = 0$

$\vec{a_2}^T(\vec{b}-\hat{b}) = 0$

$\vdots$

$\vec{a_n}^T(\vec{b}-\hat{b}) = 0$

$A^T(\vec{b}-\hat{b}) = 0$

$\vec{b}-\hat{b} \in$ null(A)

$A^T(\vec{b}-\hat{b}) = 0$

(Imp)

$A^T\vec{b} = A^T\hat{b}$ since $A\hat{x} = \hat{b}$

$= A^T(A\hat{x})$

$(A^TA)\hat{x} = A^T\vec{b}$ ← solve using LU (or) QR (or Gaussian.

$\in \mathbb{R}^{n \times n}$ $\in \mathbb{R}^n$

$\hat{x}$ = least sol!

if $\vec{x} \in \text{null}(A)$, then $\vec{x} \in$ ~~null~~ null (A^TA)

proof: Say $\vec{x} \in \text{null}(A) \Rightarrow A\vec{x} = \vec{0}$

Pre multiplying with $A^T \Rightarrow A^T(A\vec{x}) =$

$= A^T(\vec{0})$

$= \vec{0}$

$\therefore (A^TA)\vec{x} = \vec{0}$

$\Rightarrow \vec{x} \in \text{null}(A)$

If $\vec{x} \in \text{null}(A^TA)$, then $\vec{x} \in \text{null}(A)$ ⟶

↓ proof, next lecture

if Rank of $(A) = k$, $(k \leq n)$, then rank of $(A^TA) = k$

Approximate Sol ^{n} $A\vec{x} = \vec{b}$ is $\vec{x} : (A^TA)\vec{x} = A^T\vec{b}$ — (1)

Say $QR = A$

↳ Assume $\{\vec{a_1}, \dots \vec{a_n}\}$ are LI ⇒ gram schmidth will result is not $\vec{0}$

$$\vec{q_k} = a_{kk}\vec{a_k} + \sum_{i=1}^{k-1} (\vec{a_k} \cdot \vec{a_i})\vec{a_i}$$

Sub

let $A \in R^{m \times n}$ wi

col (A) =

null (A)

col (A) =

subspace of R^n

let

$A = QR \longrightarrow R^{m \times n}$

$\hookrightarrow R^{m \times n}$ ... (1)

Sub (2) in (1)

$$\left[(QR)^T QR\right] \hat{x} = [QR]^T \vec{b}$$

$$R^T Q^T Q R \hat{x} = R^T Q^T \vec{b}$$

$$\boxed{R^T R \hat{x} = R^T Q^T \vec{b}}$$

let $A \in R^{m \times n}$ with $m \geq n$:

$col(A) = \{ \vec{a_1}, \ldots \vec{a_n} \} \subseteq R^m$

$null(A) = \{ \vec{x} : A\vec{x} = 0 \} \subseteq R^n$ ($\subset$ or $\subseteq$)!!

$col(A^T) = span\{ \vec{r_1}^T, \ldots \vec{r_n}^T \} \subseteq R^n$ ($\subset$ or $\subseteq$)!

subspace of R^m

subspace of R^n

$null(A^T) = \{ \vec{y} : A^T \vec{y} = \vec{0} \} \subseteq R^m$

let $\vec{x} \in null(A)$ ...

$$A\vec{x} = \vec{0} \Rightarrow \vec{r_i} \cdot \vec{x} = 0 \quad \forall i = 1, \ldots m$$

$$\begin{bmatrix} \vec{r_1}^T \\ \vdots \\ \vec{r_m}^T \end{bmatrix} \vec{x} = 0$$

$\Rightarrow \vec{x} \perp \vec{r_i} \quad \forall i, 1 \ldots m$ since $(\vec{r_1}, \ldots \vec{r_m})$ is the spanning set of Row space of A (or) column space of (A^T)

$\Rightarrow \vec{x} \perp \text{col}(A^T) \Rightarrow$ **$\text{Null}(A) \perp \text{col}(A^T)$** $=$

$\Rightarrow$ Every vector in null space of A is $\perp$ to $\text{col}(A^T)$

$\rightarrow \dim(W) + \dim(W^T) = n$

$\rightarrow$ **$\dim(\text{Null}(A)) + \dim(\text{col}(A^T)) = n$**

$\rightarrow$ if $\vec{y} \in \text{Null}(A^T)$, then $A^T\vec{y} = \vec{0}$

$\rightarrow$

$$\begin{pmatrix} \vec{a_1} \\ \vdots \\ \vec{a_n}^T \end{pmatrix} \vec{y} = \vec{0} \quad \forall\, i = 1, \dots, n$$

$\forall$ $\vec{y} \perp \vec{a_i}$ $\forall\, i = 1, \dots, n$

$(\vec{a_1}, \dots \vec{a_n})$ is the spanning set of $\text{col}(A)$.

$\Rightarrow \vec{y} \perp \text{col}(A)$

$\Rightarrow \text{Null}(A^T) \perp \text{col}(A)$

Let $\vec{y}^T \in \text{col}(A^T) \Rightarrow \vec{y} = A^T\vec{z}$ for some $\vec{z} \in \mathbb{R}^m$

Let $\vec{x} \in \text{Null}(A) \Rightarrow A\vec{x} = \vec{0}$

$$\vec{y}^T\vec{x} = (A^T\vec{z})^T\vec{x} = \vec{z}^T A\vec{x} = \vec{z}^T\vec{0} = 0$$

$\Rightarrow \vec{y} \perp \text{Null}(A)$

$\Rightarrow$ $\dim(\text{null}(A^T)) + \dim \text{col}(A) = m$

Rank nullity theorem:

for $A \in \mathbb{R}^{m \times n}$ with $m > n$, $\underbrace{\dim \text{col}(A)}_{\text{Rank}} + \underbrace{\dim(\text{null}(A))}_{\text{nullity}} = n$

If we can show that $\dim(\text{col}(A^T)) = \dim(\text{col}(A))$, then

(2) $\rightarrow$ Rank nullity theorem.

proof:

say $(\vec{a_1}, \dots \vec{a_r})$ and LI $\Rightarrow$ $\Rightarrow \dim(\text{col}(A)) = r$

(lines independent)

basis of col(A)

(rank = col Rank)

def $\vec{v_1} = A^T \vec{a_1} \in \mathbb{R}^n$

$\vdots$

$\vec{v_r} = A^T \vec{a_r} \in \mathbb{R}^n$

say $(\vec{v_1}, \dots \vec{v_n})$ are linearly dep $\Rightarrow \exists a_i$ not all 0.

$\Rightarrow \sum_{i=1}^{r} a_i \vec{v_i} = \vec{0}$

$\Rightarrow$ $\alpha_1 A^T a_1 + \ldots + \alpha_r A^T a_r = \vec{0}$ $\quad \therefore A^T \vec{w} = 0$

$\vec{w} = \sum_{i=1}^{r} \alpha_i \vec{a_i}$

$\vec{w} \in \text{null}(A^T)$

$\vec{w} \in \text{col}(A)$ $\longleftrightarrow$ these are already I

only way $\vec{w} = 0$

$\sum_{i=1}^{r} \alpha_i \vec{a_i} = \vec{0}$ $\Rightarrow \alpha_i = 0 \Rightarrow (\vec{v_1}, \ldots, \vec{v_r})$ are linear indep

(since $\vec{a_1}, \ldots, \vec{a_r}$ are LI)

$(\vec{v_1}, \ldots, \vec{v_r})$ being LI $\Rightarrow \dim \text{col}(A^T) \geq \dim \text{col}(A) = r$ — (I)

In a similar manner, we can show that $\dim(\text{col}(A)) \geq \dim(\text{col}(A^T))$ — (II)

from (I) & (II) to hold

$$\dim(\text{col } A^T) = \dim(\text{col}(A))$$

$\Rightarrow$ row rank = col rank = r

using the above result it gets with $\dim(\text{Null}(A)) + \dim \text{col}(A^T) = n$

$\Rightarrow$ $\underbrace{\dim(\text{null}(A))}_{\text{nullity}} + \underbrace{\dim(\text{col}(A))}_{\text{rank}} = n$

$\text{rank} + \text{nullity} = n$

Trace of Matrix

- Trace! Trace is a function that operate on a square (typically) matrix & satisfies the following

$$Tr(A) : R^{n\times n} \longrightarrow R(\text{or } \mathbb{C})$$

$$Tr(A) : M_n(F) \longrightarrow F$$

properties:

a) linearity: $Tr(A+B) = Tr(A) + Tr(B)$

$$Tr(\alpha A + \beta B) = \alpha Tr(A) + \beta Tr(B) \quad \forall\, \alpha, \beta \in F \text{ and } A, B \in M_n(F)$$

b) Commutative: $Tr(AB) = Tr(BA)$

c) $Tr(I_n) = n$, where $I_n \longrightarrow n\times n$ identity matrix.

Def of Trace: $Tr(A) = \sum_{i=1}^{n} a_{ii}$ where $A \in R^{n\times n}$

d) $Tr(AB) = \sum_{i=1}^{n} (AB)_{ii} = \sum_{i=1}^{n} \langle \vec{r_i}, \vec{b_i} \rangle$

$$Tr(BA) = \sum_{i=1}^{n} (BA)_{ii}$$

Say. $A = \begin{bmatrix} \vec{r_1} \\ \\ \vec{r_n} \end{bmatrix}$ & $B = [\vec{b} \cdots \vec{b_n}]$

b) $\mathrm{Tr}(A^TA) = \sum_{i=1}^{n} \langle \vec{a_i}, \vec{a_i} \rangle = \sum_{i=1}^{n} \|\vec{a_i}\|^2$

Frobenious norm of a matrix: $\|A\|_F = \sqrt{\mathrm{Tr}(A^TA)}$

c) $\mathrm{Tr}(\vec{u}\vec{u}^T) = \begin{bmatrix} u_1\vec{u}^T \\ \vdots \\ u_n\vec{u}^T \end{bmatrix} \Rightarrow \mathrm{Tr}(\vec{u}\vec{u}^T) = \sum_{i=1}^{n} u_i^2$

$= \langle \vec{u}, \vec{u} \rangle$

check for

$\Rightarrow \mathrm{Tr}(ABC) = \mathrm{Tr}(BCA) = \mathrm{Tr}(CAB)$

Determinant of a matrix

det: Determinant is a function that satisfies.

↳ $\mathrm{Det}(A): M_n(F) \to F(\text{or } \mathbb{R}^{n\times n} \to \mathbb{R})$

$A = [\vec{a_1} \cdots \vec{a_n}]$

a) multilinearity.

$\det[\vec{a_1} \cdots \vec{a_n}])$ with $\vec{a_i} = \alpha\vec{u} + \beta\vec{v}$

$= \alpha \det[\vec{a_1} \cdots \vec{u}, \cdots \vec{a_n}]$

$+ \beta \det[\vec{a_1}, \cdots \vec{v}, \cdots \vec{a_n}]$

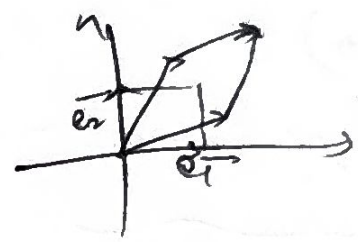
b) swap any two columns.

$\det[\vec{a_1}, \cdots \vec{a_i}, \cdots \vec{a_j}, \cdots \vec{a_n}]$

$= -\det[\vec{a_1} \cdots \vec{a_j}, \cdots \vec{a_i} \cdots \vec{a_n}]$

c) $\det(I_n) = 1$

ex. $\det\begin{bmatrix} a & b \\ c & d \end{bmatrix} =$



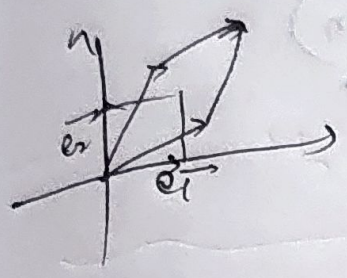
Derived:

2) det of a

3) Det of upp

ex. $\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = a \det(\vec{e_1}, a\vec{e_2}) + c \det(\vec{e_1}, a\vec{e_2})$

$= a\, b\det(\vec{e_1}, b\vec{e_1} + d\vec{e_2}) + c \det(\vec{e_1}, b\vec{e_1} + d\vec{e_2})$



$= ab \det(\vec{e_1}, \vec{e_1}) + ad \det(\vec{e_1}, \vec{e_2}) + bc \det(\vec{e_2}, \vec{e_1}) + cd \det(\vec{e_1}, \vec{e_2})$

$= ad - bc$

Derived: 1) Det of a matrix with same columns.

(say $\vec{a_i} = \vec{a_j}$)

$\det(\vec{a_1}, \dots \vec{a_i} \dots \vec{a_j} \dots \vec{a_n}) = -\det(\vec{a_1} \dots \vec{a_j}, -\vec{a_i} \dots \vec{a_n})$

$\Rightarrow 0$

2) det of a matrix with linearly dep column vectors.

say $\vec{a_k} = \alpha \vec{a_i} + \beta \vec{a_j}$

by multilinearity $\det(\vec{a_1} \dots \vec{a_n}) = \alpha \det(\vec{a_1} \dots \vec{a_i}, \dots \vec{a_j}, \dots \vec{a_n}) + \beta \det(\vec{a_1} \dots \vec{a_i}, \dots \vec{a_j}, \dots \vec{a_n})$

$= 0 + \beta \cdot 0$

$= 0$

3) Det of upper Δ le matrix = product of the diagonal entries.

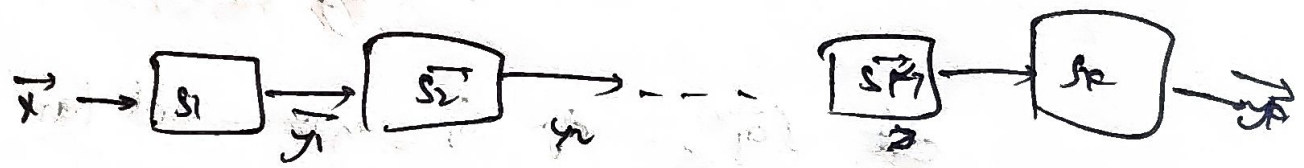
Eigenvalues and Eigenvectors



$$\vec{y} = A\vec{x}$$

↖ transformation matrix..

Q: what is the o/p when we connect $\vec{x}$ such systems in cascade manner.



$$\vec{y_k} = A\vec{y}_{k-1}$$

$$= A^2\vec{y}_{k-2}$$

$$\boxed{\vec{y_k} = A^k\vec{x}}$$

↳ other scenarios - except when such eq's. are encountered

→ Given

a) what is $\vec{y_k}$?

b) what happens to $\vec{y_k}$ for $k \to \infty$.

Scen

Scenario 1: $\vec{x}$ is a space vector : $A\vec{x} = \lambda\vec{x}$

↓

where $\vec{x}$ is that vector whose direction does not change.

λ → scalar
A → matrix.

$$y_2 = A\vec{x}$$
$$= A(A\vec{x})$$
$$= A(\lambda\vec{x})$$
$$= \lambda A\vec{x}$$
$$= \lambda^2\vec{x}$$

$$y_2 = \lambda^2\vec{x}$$

$$y_k = \lambda^k\vec{x} \rightarrow \text{output is in the same direction.}$$

Norm of y^k $\Rightarrow$ $\|y_k\| = |\lambda^k|\|\vec{x}\| = |\lambda|^k\|\vec{x}\|$

if $|\lambda| = 1 \rightarrow \|\vec{y}^k\| = \|\vec{x}\|$ as $k \to \infty$

if $|\lambda| < 1 \rightarrow \|\vec{y}^k\| = 0$ as $k \to \infty$

if $|\lambda| > 1 \rightarrow \|\vec{y}^k\| \to \infty$ as $k \to \infty$

→ given A; how many $\vec{x}$ exist s.t. $A\vec{x} = \lambda\vec{x}$.

Scenario 2: say $\exists$ $(\vec{v_1}, \ldots \vec{v_n})$ s.t. $A\vec{v_1} = \lambda \cdot \vec{v_1}$

and $(\vec{v_1}', \ldots \vec{v_n}')$ and so on.

(span)

any $\vec{x} = \sum_{i=1}^{n} \alpha_i \vec{v_i}$

$y_1 = Ax = A\left(\sum_{i=1}^{n} \alpha_i v_i\right) = \sum_{i=1}^{n} \alpha_i (A v_i)$

$= \sum_{i=1}^{n} \alpha_i \lambda_i x$

$y_k = A^k x = \sum_{i=1}^{n} \alpha_i \lambda_i^k v_i$

$= \alpha_1 \lambda_1^k v_1 + \cdots + \alpha_n \lambda_n^k v_n$

arrange arrange in descending order.

Indexing: $|\lambda_1| \geq |\lambda_2| \geq |\lambda_3| \geq \cdots \geq |\lambda_n|$

$y_k = \lambda_1^k \left(\alpha_1 v_1 + \sum_{i=2}^{n} \left(\frac{\lambda_i}{\lambda_1}\right)^k v_i\right)$

$(\alpha_1 \neq 0)$

output

$y_k \approx \lambda_1^k \alpha_1 v_1 \rightarrow$ along v_1 as $k \to \infty$

if $\lambda_1 < 1$ $\|y_k\| \to 0$ as $k \to \infty$

if $|\lambda_1| > 1$ $\|y_k\| \to \infty$ as $k \to \infty$

let $V = [v_1 \cdots v_n]$

$AV = A[v_1 \cdots v_n]$

$= [\lambda_1 v_1 \cdots \lambda_n v_n] = V\Lambda$

$= VD$

Λ = Diagonal matrix

$D = (\lambda_1 \cdots \lambda_n)$

$AV = VD$

$\Rightarrow AVV^{-1} = VDV^{-1}$

V^{-1} exists since rank is n (linearly indep)

$\boxed{A = VDV^{-1}}$

↳ if A^T is symmetric, $V^{-1} = V^T$ (not true vice versa)

↓

when symmetric $A^k = A \cdots A$

$= (VDV^T) \cdots$

$= VD^kV^T$

↳ $diag(\lambda_1^k, \cdots \lambda_n^k)$

if not symmetry

$A^k = A \cdots A$

$= (VDV^{-1})(VDV^{-1}) \cdots$

$= VD^kV^T$

↳ $Diag(\lambda^k, \cdots \lambda n^k)$

$\boxed{e^{At} = VAV^T \quad \text{where } A = diag(e^{\lambda_1 t}, \cdots e^{\lambda_n t})}$

eigenvalue & eigenvector:

eigen value & eigenvector of A: λ is an eigen value of λ ; If $\exists \vec{v}$ s.t

$$A\vec{v} = \lambda\vec{v}$$

$\lambda \leftarrow$ eigenvalue

$\vec{v} \leftarrow$ eigen vector associated with λ.

To compute λ & $\vec{v}$: $A\vec{v} = \lambda\vec{v} \Rightarrow (A-\lambda I)\vec{v} = 0$

$\Rightarrow \vec{v} \in (\text{null } (A-\lambda I)$

$\Rightarrow \vec{v} \neq 0$ if $\text{rank}(A-\lambda I) < n$

$\det(A-\lambda I) = 0$

↓

polynomial of nth order in λ.

↓

characteristic polynomial.

↓

Solving this we get λ.

↓

maximum. of n such λ's.

(n eigenvalues)

↓

Can be real, complex, repeated

are eigenval ~~are~~ - compl: $A\vec{v_i} = \lambda_i \vec{v_i}$

eigen space : associated with λ_i = collection of all eigenvectors $\vec{v_i}$

↳ subspace of R^n.

$\vec{V} = A\vec{v} = \lambda\vec{v}$

ex: $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ $\therefore$ $P_\lambda = \begin{vmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{vmatrix}$

$= \lambda^2 - 4\lambda - 4\lambda + 4$

$= \lambda^2 - 4\lambda + 3 = 0$

$= \therefore \lambda = 3, 1$

$\lambda_1 = 3 \;,\; \lambda_2 = 1$

$(A - \lambda_1 I)\vec{v_1} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} v_1 \end{bmatrix} = 0$

$-v_{11} + v_{12} = 0$

$v_{11} - v_{12} = 0$

$\because v_{12} = 1$

$\therefore v_{11} = 1$

$\vec{v_1} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ $\vec{q_1} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$

Sum & product of eigen values:

Characteristic polynomial of A $P_A(\lambda) = \det(A - \lambda I)$

↓

Polynomial of order n.

↓

n roots (real or complex (or repeat))

Assume $\lambda_1, \dots, \lambda_m, \lambda_{m+1}, \dots, \lambda_{m+1}$ — m+1 distinct eig values

λ_{m+1} repeat r times ⟵ $r = n - (m+1)$ repeated eig values

$\det(\lambda I - A)$

$\det(A - \lambda I) = (\lambda - \lambda_1) \cdots (\lambda - \lambda_m)(\lambda - \lambda_{m+1})^{r+1}$

when $\lambda = 0$ ⇒ $\det(A) = (-\lambda_1) \cdots (-\lambda_m)(-\lambda_{m+1})^{r+1}$

$(-1)^n \det(A) = (-1)^n$ (prod. of eig values)

$$\Rightarrow \boxed{\det(A) = \prod_{i=1}^{n} \lambda_i}$$

ex: $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$; $P_A(\lambda) = (2-\lambda)(2-\lambda) - 1$

$= \lambda^2 + 3 - 4\lambda$

$= (\lambda - 3)(\lambda - 1)$

ex $A = \text{diag}(2, 1, 2)$ ⇒ $P_A(\lambda) = \det(A - \lambda I) = (2-\lambda)^2 (1-\lambda) = -\lambda^3 + \dots$

$P_A(\lambda) = \det(\lambda I - A) = (\lambda - 2)^2 (\lambda - 1)$

result (without proof): $\boxed{\text{Tr}(A) = \text{sum of eig values}}$

(Vieta's equation)

a) There is only one eigen value associated with any eigen vector.

any eigen vector belongs to a unique eigenspace(E)

proof: (Sagar) if $\lambda_1 \neq \lambda_2$ exist such that ~~$\vec{v_1} = \vec{v_2}$~~

$$A\vec{v} = \lambda_1\vec{v} \quad \text{&}$$

$$A\vec{v} = \lambda_2\vec{v}$$

$$0 = (\lambda_1 - \lambda_2)\vec{v}$$

$$\therefore \boxed{\lambda_1 = \lambda_2}$$

(∵ matrix vector product is defined uniquely.

b) Eigen vectors corresponding to distinct eigenvalues are linearly independent

proof: Say $\exists\ \vec{v_1} \in V_2$ whose eigenvalues are $\lambda_1 \neq \lambda_2$ respectively. ie $A\vec{v_1} = \lambda_1\vec{v_1}$ &

$$A\vec{v_2} = \lambda_2\vec{v_2}$$

Say $\vec{v_2} = \alpha\vec{v_1}$ (assume that $\vec{v_1}$ & $\vec{v_2}$ are linearly dependent)

pre multiply (2) with A

$$\Rightarrow \quad A\vec{v_2} = A(\alpha \vec{v_1})$$

$$\lambda_2 \vec{v_2} = \alpha(\lambda_1 \vec{v_1})$$

$$= \alpha \frac{\lambda_1}{\lambda_2} \vec{v_1} \quad - (3)$$

(2) & (3) hold only if $\lambda_1 = \lambda_2$ → eg values are same.

↓

Contradiction

c) If λ is a complex eigenvalue of $A \in \mathbb{R}^{n \times n}$ with eigenvalue with eigenvector v.

eigenvector v, then λ^* is also an eigen value with eigen vector v^*.

proof:- since λ is a complex eigenvalue. with eigen vector $\vec{v}$ complex.

$$A\vec{v} = \lambda \vec{v}$$

$$(A\vec{v})^* = (\lambda \vec{v})^*$$

$$A\vec{v}^* = \lambda^* \vec{v}^*$$

$\Rightarrow \lambda^*$ is the eigen value with eigen vector v^*.

⇒ eigen values of complex appear is conjugat pairs.

d) If A is a real symmetric matrix, then all the eigen values are real.

Proof:

A is Symmetric $\Rightarrow A^T = A$

$\vec{v}$ is an eigen vector of A with the eigen value $\lambda_1 \Rightarrow A\vec{v} = \lambda\vec{v}$

$\Rightarrow$ inner product of any vector is real.

$$\left((A\vec{v})^{*T}(A\vec{v})\right) \in R$$

$$(\vec{v}^{*})^T (A^{*})^T A\vec{v} = (\lambda^{*}\vec{v}^{*})^T (\lambda\vec{v})$$

$$\downarrow$$

$$A^T = A$$

$$(\vec{v}^{*})^T A^2 \vec{v} \in R$$

$$\& \quad (\vec{v}^{*})^T A^2 \vec{v} = \lambda^2 (\vec{v}^{*})^T \vec{v}$$

$$\|A\vec{v}\|^2 = \|\vec{v}\|^2$$

$$\Rightarrow \lambda^2 = \frac{\|A\vec{v}\|^2}{\|\vec{v}\|^2} = \in R$$

$|\lambda| \in R$

$\Rightarrow$ $\boxed{\lambda = \text{real}}$

e) If A is real symmetric matrix, then eigen vectors corresponding to distinct eigenvalues are orthogonal.

proof: Say that $\vec{v_1}$ & $\vec{v_2}$ are eigen vectors of A (symmetric) with eigen values λ_1 & λ_2

$\vec{v_1}^T \vec{v_2} = 0$

$$(A\vec{v_1})^T (A\vec{v_2}) = \vec{v_1}^T A^T A \vec{v_2}$$

$$= \vec{v_1}^T A^2 \vec{v_2}$$

$(A\vec{v_2})$ $A\lambda_1$